An Inverse Riccati Construction of Second-Order Linear ODEs from Degree-2 Algebraic Extensions

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Abstract

We present an explicit two-pole construction of a second-order linear ODE of the form $y'' + r(z)y = 0$ whose solutions are Liouvillian and arise from a logarithmic derivative $s(z) \in \mathbb{C}(z, \sqrt{R})$ of algebraic degree exactly 2 over $\mathbb{C}(z)$. The construction places two poles at $z = 0$ and $z = 1$ with equal residues 1/4, yielding the potential $r(z) = \frac{3}{16z^2(z-1)^2}$ and solutions expressible in terms of radicals and inverse hyperbolic functions. We verify the Riccati equation by direct computation, prove that the logarithmic derivative has algebraic degree 2 via its minimal polynomial, and establish linear independence of the two fundamental solutions via their nonconstant ratio and constant Wronskian. An outline of an n-pole generalization with a complete irrational cancellation identity is included. We have not found this exact inverse two-pole construction in the standard references consulted.

1 Introduction

The Kovacic algorithm [1] provides a systematic method for determining whether a second-order linear ODE of the form

$$y'' + r(z)y = 0 \tag{1}$$

has Liouvillian solutions, i.e., solutions expressible in terms of exponentials, logarithms, and algebraic functions. The algorithm operates by examining the poles of the coefficient function $r(z)$ and classifying the local behavior at each singular point into one of four cases, each of which constrains the possible algebraic degree of the logarithmic derivative of a solution.

A Liouvillian solution of (1) can be written as $y_1 = \exp(\int s(z)\,dz)$, where $s(z) \in \mathbb{C}(z, \sqrt{R})$ satisfies the Riccati equation

$$s' + s^2 + r(z) = 0. \tag{2}$$

The algebraic degree of $s(z)$ over $\mathbb{C}(z)$ determines the complexity of the solution. Kovacic's algorithm separates the problem into cases according to the possible structure of an algebraic Riccati solution. In this paper we focus on the quadratic case, where $s(z) \in \mathbb{C}(z, \sqrt{R})$ for some square-free $R \in \mathbb{C}(z)$ with $\sqrt{R} \notin \mathbb{C}(z)$.

We present a construction that starts from the desired algebraic structure—a quadratic extension $\mathbb{C}(z, \sqrt{R})/\mathbb{C}(z)$—and derives the ODE coefficient $r(z)$ from the Riccati equation. This is the reverse of the usual approach, which starts from $r(z)$ and attempts to find $s(z)$. Our construction naturally produces ODEs with two poles, in contrast to many of the single-pole or single-point examples we encountered in the references consulted within Kovacic's framework.

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2 The Two-Pole Ansatz

We seek a logarithmic derivative of the form

$$s(z) = A(z) + B(z)\sqrt{R(z)}, \quad A, B, R \in \mathbb{C}(z), \quad \sqrt{R} \notin \mathbb{C}(z), \tag{3}$$

where $s(z)$ lies in the quadratic extension $\mathbb{C}(z, \sqrt{R})$ of $\mathbb{C}(z)$. The Riccati equation implies $r(z) = -(s' + s^2)$.

We place two poles at $z = 0$ and $z = 1$ with equal residues $1/4$. This gives

$$A(z) = \frac{2z-1}{4z(z-1)} = \frac{1}{4z} + \frac{1}{4(z-1)}, \tag{4}$$

$$R(z) = \frac{1}{z(z-1)}, \tag{5}$$

$$B = c = \frac{1}{2}. \tag{6}$$

For the explicit two-pole construction we take $B = c = 1/2$, a constant. In this constant-B case, the irrational part of $s' + s^2$ vanishes when

$$\frac{R'}{2} + 2AR = 0, \tag{7}$$

which is verified by direct computation in Theorem 1. The resulting potential depends on c^2; the choice $c = 1/2$ gives the simple potential used below. The nonconstant-B cancellation condition and its n-pole parametrization are developed in Section 7. This yields the two branches

$$s_\pm(z) = \frac{2z-1}{4z(z-1)} \pm \frac{1}{2\sqrt{z(z-1)}}. \tag{8}$$

The two poles of A are located at $z = 0$ and $z = 1$, each with residue $1/4$; the branch cut for $\sqrt{z(z-1)}$ is taken along $[0, 1]$.

2.1 Field-Extension Tower

$$\begin{array}{c} \mathbb{C}(z, \sqrt{R}) \quad \text{[degree 2]} \\ \uparrow{\scriptstyle \sqrt{R}\notin\mathbb{C}(z)} \\ \mathbb{C}(z) \quad \text{[rational functions]} \end{array}$$

Quadratic extension for $s(z)$.

3 The Construction

3.1 Choosing the Poles and Computing A, B, R, s

We choose poles at $z_1 = 0$ and $z_2 = 1$ with residues $a_1 = a_2 = 1/4$ (corresponding to $n_1 = n_2 = 1$, both odd). The partial-fraction decomposition of A is

$$A(z) = \frac{2z-1}{4z(z-1)} = \frac{1}{4z} + \frac{1}{4(z-1)}.$$

The radicand R encodes the irrational part of the logarithmic derivative:

$$R(z) = \frac{1}{z(z-1)}.$$

With $B = c = 1/2$ (the simplest subfamily, $f(z) = 1$), the two branches of the logarithmic derivative are

$$s_\pm(z) = \frac{2z-1}{4z(z-1)} \pm \frac{1}{2\sqrt{z(z-1)}}.$$

4 The ODE and Its Solutions

4.1 The Potential $r(z)$

From the Riccati equation, $r(z) = -(s' + s^2)$. We first compute the rational part $A' + A^2 + B^2R$. The irrational part is shown below to vanish for this choice of A and R:

$$A' = -\frac{1}{4z^2} - \frac{1}{4(z-1)^2}, \tag{9}$$

$$A^2 = \frac{1}{16z^2} + \frac{1}{8z(z-1)} + \frac{1}{16(z-1)^2}, \tag{10}$$

$$B^2R = \frac{1}{4z(z-1)}. \tag{11}$$

Summing these three contributions,

$$A' + A^2 + B^2R = -\frac{3}{16z^2} - \frac{3}{16(z-1)^2} + \frac{3}{8z(z-1)} = -\frac{3}{16z^2(z-1)^2}. \tag{12}$$

Therefore

$$r(z) = -(A' + A^2 + B^2R) = \frac{3}{16z^2(z-1)^2}. \tag{13}$$

The irrational parts cancel because $R'/2 + 2AR = 0$ for our specific A and R, as verified explicitly in Theorem 1. Thus both branches yield the same $r(z) \in \mathbb{C}(z)$.

4.2 Integrating s to Obtain y_1

The first solution is

$$y_1 = \exp\left(\int s_+ \, dz\right). \tag{14}$$

We split the integral:

$$\int s_+ \, dz = \int \frac{2z-1}{4z(z-1)} \, dz + \int \frac{1}{2\sqrt{z(z-1)}} \, dz. \tag{15}$$

The first integral is elementary, and the second is expressed using arcosh:

$$\int s_+ \, dz = \frac{1}{4} \log(z(z-1)) + \frac{1}{2} \operatorname{arcosh}(2z-1). \tag{16}$$

Using the complex identity $\operatorname{arcosh}(2z-1) = \log(2z - 1 + 2\sqrt{z(z-1)})$, valid after choosing compatible branches, we obtain $\int s_+ \, dz = \frac{1}{4} \log(z(z-1)) + \frac{1}{2} \log(2z - 1 + 2\sqrt{z(z-1)})$. Exponentiating gives

$$y_1 = [z(z-1)]^{1/4} [2z - 1 + 2\sqrt{z(z-1)}]^{1/2}. \tag{17}$$

4.3 The Second Solution y_2 from the Conjugate

The second solution arises from the conjugate logarithmic derivative

$$s_2 = A - B\sqrt{R} \tag{18}$$

obtained by applying the nontrivial automorphism of the quadratic extension $\mathbb{C}(z, \sqrt{R})/\mathbb{C}(z)$ which sends $\sqrt{R} \mapsto -\sqrt{R}$. This yields

$$y_2 = [z(z-1)]^{1/4} [2z - 1 - 2\sqrt{z(z-1)}]^{1/2}. \tag{19}$$

The automorphism preserves the Riccati equation because $r(z) \in \mathbb{C}(z)$ is fixed, so s_2 is also a root and $y_2 = \exp(\int s_2 \, dz)$ is also a solution.

5 Proofs

Theorem 1 (Riccati Verification)**.** *Let* $A(z) = \frac{2z-1}{4z(z-1)}$*,* $B = \frac{1}{2}$*,* $R(z) = \frac{1}{z(z-1)}$*, and* $s_\pm = A \pm B\sqrt{R}$*. Define* $r(z) = -(A' + A^2 + B^2R)$*. Then*

$$r(z) = \frac{3}{16z^2(z-1)^2}$$

and both s_+ *and* s_- *satisfy the Riccati equation* $s' + s^2 + r(z) = 0$*.*

Proof. The rational computation is given in (13). It remains to verify that the irrational part vanishes. For $s_+ = A + B\sqrt{R}$,

$$s_+' + s_+^2 = A' + A^2 + B^2R + \left(\frac{BR'}{2\sqrt{R}} + 2AB\sqrt{R}\right).$$

The irrational part is $\frac{B}{\sqrt{R}}\left(\frac{R'}{2} + 2AR\right)$. For our specific A and R,

$$\frac{R'}{2} + 2AR = -\frac{2z-1}{2z^2(z-1)^2} + \frac{2(2z-1)}{4z(z-1)} \cdot \frac{1}{z(z-1)} = -\frac{2z-1}{2z^2(z-1)^2} + \frac{2z-1}{2z^2(z-1)^2} = 0.$$

Thus the irrational part vanishes, and $s_+' + s_+^2 = A' + A^2 + B^2R = -r(z)$. The case $s_- = A - B\sqrt{R}$ is identical with B replaced by $-B$; since the irrational term is proportional to B and equals zero, both branches satisfy the Riccati equation. □

Theorem 2 (Algebraic Degree 2)**.** *The logarithmic derivative* $s = A + B\sqrt{R}$ *has algebraic degree exactly 2 over* $\mathbb{C}(z)$*; that is,* $s \notin \mathbb{C}(z)$ *and* s *satisfies a degree-2 minimal polynomial over* $\mathbb{C}(z)$*.*

Proof. Since $s = A + B\sqrt{R}$ with $B \neq 0$, the minimal polynomial is obtained by eliminating $\sqrt{R}$: $(s - A)^2 = B^2R$, or

$$s^2 - 2As + (A^2 - B^2R) = 0. \tag{20}$$

Substituting $A = (2z-1)/(4z(z-1))$, $B = 1/2$, and $R = 1/(z(z-1))$ gives:

$$s^2 - \frac{2z-1}{2z(z-1)}s + \frac{1}{16z^2(z-1)^2} = 0. \tag{21}$$

Here $R = 1/(z(z-1))$ is not a square in $\mathbb{C}(z)$, since it has simple poles at $z = 0$ and $z = 1$, while a square in $\mathbb{C}(z)$ can only have even-order poles. Thus $\sqrt{R} \notin \mathbb{C}(z)$; since $B \neq 0$, $s \notin \mathbb{C}(z)$. The polynomial (21) has degree 2 in s with coefficients in $\mathbb{C}(z)$. Since $s \notin \mathbb{C}(z)$, it is irreducible over $\mathbb{C}(z)$ and hence is the minimal polynomial. □

Theorem 3 (Linear Independence)**.** *The solutions* y_1 *and* y_2 *are linearly independent over* $\mathbb{C}$*.*

Proof. Set $u = 2z - 1 + 2\sqrt{z(z-1)}$ and $v = 2z - 1 - 2\sqrt{z(z-1)}$. The identity $(2z-1)^2 - 4z(z-1) = 1$ gives $uv = 1$. On any branch where y_1 and y_2 are defined, their ratio is

$$\frac{y_1}{y_2} = \left(\frac{u}{v}\right)^{1/2}.$$

Since $v = 1/u$, this ratio is $\pm u$, depending on branch choices. But $u = 2z - 1 + 2\sqrt{z(z-1)}$ is a nonconstant algebraic function. Therefore y_1/y_2 is not constant on any open domain where the branches are defined, and y_1, y_2 are linearly independent.

Additionally, by Abel's identity, the Wronskian is constant because the ODE has no y' term. Direct evaluation gives $W(y_1, y_2) = \pm 1$, which is constant and nonzero on each branch. □

6 Computational Verification

The complete executable verification script is available as `code/verify.py` in the accompanying repository. It includes assertions for all checks below and is run automatically via GitHub Actions on every push. The essential checks are:

Listing 1: SymPy Verification (excerpt)

```python
import sympy as sp
z = sp.Symbol('z')

# Define A, B, R
A = (2*z - 1) / (4*z*(z - 1))
B = sp.Rational(1, 2)
R = 1 / (z*(z - 1))
s_plus = A + B * sp.sqrt(R)
s_minus = A - B * sp.sqrt(R)

# r(z) = -(A' + A^2 + B^2*R)
r = -(sp.diff(A, z) + A**2 + B**2*R)
assert sp.simplify(r - sp.Rational(3,16)/(z**2*(z-1)**2)) == 0

# Riccati: s' + s^2 + r = 0 for both branches
assert sp.simplify(sp.diff(s_plus, z) + s_plus**2 + r) == 0
assert sp.simplify(sp.diff(s_minus, z) + s_minus**2 + r) == 0

# Minimal polynomial is quadratic in s
s = sp.Symbol('s')
assert sp.Poly(sp.simplify(s**2 - 2*A*s + (A**2 - B**2*R)), s).degree() == 2

# Key identity: (2z-1)^2 - 4z(z-1) = 1
assert sp.expand((2*z - 1)**2 - 4*z*(z - 1)) == 1

# ODE check: y'' + r*y = 0
y1 = (z*(z-1))**sp.Rational(1,4) * (2*z-1 + 2*sp.sqrt(z*(z-1)))**sp.Rational(1,2)
y2 = (z*(z-1))**sp.Rational(1,4) * (2*z-1 - 2*sp.sqrt(z*(z-1)))**sp.Rational(1,2)
assert sp.simplify(sp.diff(y1, z, 2) + r*y1) == 0
assert sp.simplify(sp.diff(y2, z, 2) + r*y2) == 0

# Wronskian: W^2 = 1 (branch-aware)
assert sp.simplify((z*(z-1)) * (s_minus - s_plus)**2) == 1

# Ratio nonconstancy: u' != 0
u = 2*z - 1 + 2*sp.sqrt(z*(z-1))
assert not sp.simplify(sp.diff(u, z)).equals(0)
```

All computational checks pass. The SymPy verification is consistent with the formal proofs in Section 5. Note that exact algebraic degree 2 (irreducibility of the minimal polynomial) relies on the proof that R is not a square in $\mathbb{C}(z)$, established in Theorem 2.

7 Outline of an n-Pole Generalization

The construction extends formally to n poles at pairwise distinct locations $z_1, \dots, z_n \in \mathbb{C}$ with residues $a_i = n_i/4$, where each n_i is odd. The framework is

$$A(z) = \sum_{i=1}^{n} \frac{a_i}{z - z_i}, \quad a_i = \frac{n_i}{4}, \quad n_i \in \{1, 3, 5, \dots\} \tag{22}$$

$$\exp\left(-2 \int A\, dz\right) = \prod_{i=1}^{n} (z - z_i)^{-2a_i} = \prod_{i=1}^{n} (z - z_i)^{-n_i/2}. \tag{23}$$

The square root of R is then

$$\sqrt{R} = \frac{\prod_{i=1}^{n} (z - z_i)^{-n_i/2}}{f(z)}, \quad f(z) \in \mathbb{C}(z). \tag{24}$$

The radicand and coefficient B are

$$R = \frac{\prod_{i=1}^{n} (z - z_i)^{-n_i}}{f(z)^2} \tag{25}$$

$$B = cf(z). \tag{26}$$

For a general rational coefficient $B(z)$, the irrational part of $s' + s^2$, with $s = A + B\sqrt{R}$, vanishes precisely when

$$\frac{B'}{B} + \frac{R'}{2R} + 2A = 0. \tag{27}$$

Taking

$$B = cf(z), \qquad R = \frac{\prod_{i=1}^{n} (z - z_i)^{-n_i}}{f(z)^2} \tag{28}$$

satisfies this identity identically, because

$$\frac{B'}{B} = \frac{f'}{f}, \qquad \frac{R'}{2R} = -2A - \frac{f'}{f}, \tag{29}$$

where the second identity follows from logarithmic differentiation of $R = \prod_i (z - z_i)^{-n_i}/f(z)^2$. Therefore

$$\frac{B'}{B} + \frac{R'}{2R} + 2A = \frac{f'}{f} + \left(-2A - \frac{f'}{f}\right) + 2A = 0. \tag{30}$$

In the simplest subfamily $f(z) = 1$, one has $B = c$, a constant, recovering the two-pole example treated above.

Note that $f(z)$ is auxiliary: it cancels from both $B\sqrt{R}$ and B^2R, since

$$B\sqrt{R} = cf(z) \cdot \frac{\prod_{i=1}^{n} (z - z_i)^{-n_i/2}}{f(z)} = c \prod_{i=1}^{n} (z - z_i)^{-n_i/2}, \tag{31}$$

$$B^2R = c^2 f(z)^2 \cdot \frac{\prod_{i=1}^{n} (z - z_i)^{-n_i}}{f(z)^2} = c^2 \prod_{i=1}^{n} (z - z_i)^{-n_i}. \tag{32}$$

Thus this parametrization does not produce distinct ODE coefficients as f varies; the essential parameters are the pole locations z_i, the odd integers n_i, and the constant c.

The ODE coefficient is $r(z) = -(A' + A^2 + B^2R)$, which depends on c^2 through the B^2R term. For $c \in \mathbb{C}^\times$, the logarithmic derivatives lie in a quadratic extension and have degree exactly 2; the value $c = 0$ is a degenerate rational case with $s = A \in \mathbb{C}(z)$. Because the z_i are distinct and each n_i is odd, R has odd valuation at every z_i and therefore is not a square in

$\mathbb{C}(z)$, which ensures the degree is exactly 2 for $c \neq 0$. For the two-pole case with general c, this yields the one-parameter family

$$r_c(z) = \frac{3}{16z^2} + \frac{3}{16(z-1)^2} - \frac{1/8 + c^2}{z(z-1)} = \frac{4(1-4c^2)z(z-1)+3}{16z^2(z-1)^2}. \tag{33}$$

For $c = 1/2$, the $z(z-1)$ term vanishes and one recovers $r(z) = \frac{3}{16z^2(z-1)^2}$. Since the potential depends only on c^2, the parameters c and $-c$ determine the same ODE and merely interchange the two conjugate Riccati branches.

8 Comparison with Kovacic's Examples

Kovacic's original paper [1] and subsequent literature, such as [2, 3], study ODEs with a single irregular singular point or specific pole configurations arising from special-function theory, including Bessel, Airy, and hypergeometric examples. Our construction differs from examples we encountered in the consulted references in the following sense:

Feature	**Examples encountered in consulted references**	**This construction**
Pole structure	Single or multi-pole configurations	Two distinct poles at $z = 0$ and $z = 1$
Explicit $r(z)$	Varies, e.g., $\frac{1}{4z^2}$, $\frac{-1+3z^2}{4z^2(z^2-1)^2}$	$\frac{3}{16z^2(z-1)^2}$
Algebraic extension	Examples such as $\mathbb{C}(z, \sqrt{az+b})$	$\mathbb{C}(z, \sqrt{1/(z(z-1))})$
Construction direction	Given $r(z)$, find $s(z)$	Given the structure of $s(z)$, derive $r(z)$
Residue pattern	Various	Equal residues 1/4 at each pole

In the examples from Kovacic and the references we consulted [1, 2, 3, 4], we did not find this exact inverse two-pole construction. The construction is natural from the perspective of inverse Riccati problems but appears to be less common in the examples we consulted.

9 Conclusion

We have presented a systematic construction of second-order linear ODEs with degree-2 Liouvillian solutions, based on a two-pole ansatz for the logarithmic derivative $s(z) = A(z) + B(z)\sqrt{R(z)}$. The specific example with poles at $z = 0$ and $z = 1$ yields the ODE $y'' + \frac{3}{16z^2(z-1)^2}y = 0$ with solutions

$$y_1 = [z(z-1)]^{1/4}[2z - 1 + 2\sqrt{z(z-1)}]^{1/2}$$
$$y_2 = [z(z-1)]^{1/4}[2z - 1 - 2\sqrt{z(z-1)}]^{1/2}$$

The construction was verified by: (i) direct computation of $r(z)$ from $-(A' + A^2 + B^2R)$; (ii) proof that s has algebraic degree exactly 2 via its minimal polynomial; (iii) proof of linear independence via the nonconstant ratio y_1/y_2 and constant nonzero Wronskian $W = \pm 1$; and (iv) supporting SymPy computational checks.

The n-pole generalization extends the framework to arbitrary numbers of poles with odd residue parameters. The auxiliary function $f(z)$ enters the parametrization but cancels from the resulting ODE coefficient; the essential parameters are the pole locations, the odd integers n_i, and the constant c. Further exploration of the resulting multi-pole ODE families is a natural direction for future work.

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